

♦ **Basics of Azure Data Lake (ADLS Gen1 & Gen2)**

1. What is Azure Data Lake and what are its primary use cases?
 2. Differentiate between Azure Data Lake Gen1 and Gen2.
 3. What are the key components of Azure Data Lake?
 4. How does ADLS integrate with Azure Storage?
 5. What is the importance of hierarchical namespace in ADLS Gen2?
 6. What is the difference between Blob Storage and ADLS Gen2?
 7. How is data organized in Azure Data Lake?
 8. What are the benefits of using ADLS over traditional HDFS?
 9. What types of data can you store in ADLS?
 10. Explain the lifecycle of a file in ADLS.
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♦ **Security and Access Control**

11. How do you secure data in Azure Data Lake?
12. What is Access Control List (ACL) in ADLS Gen2?
13. How does Azure RBAC differ from ACLs in ADLS?
14. What is the least privilege model in ADLS?
15. How can you implement multi-level security in ADLS?
16. What is the role of Azure Active Directory in ADLS?
17. How do you grant access to external users in ADLS?
18. Explain how OAuth 2.0 is used for authentication in ADLS.

19. How do you monitor and audit access to data in ADLS?
 20. What are best practices for managing permissions in ADLS?
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◆ **Data Ingestion & Integration**

21. What are the different ways to ingest data into ADLS?
 22. How do you use Azure Data Factory with ADLS?
 23. How does Azure Databricks integrate with ADLS?
 24. Can you ingest streaming data into ADLS? If yes, how?
 25. How do you perform incremental loads into ADLS?
 26. What is PolyBase and how does it work with ADLS?
 27. Describe the role of Azure Synapse in ADLS data integration.
 28. How would you ingest data from on-premise to ADLS?
 29. Explain the use of Logic Apps in data ingestion to ADLS.
 30. How do you ingest data from REST APIs into ADLS?
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◆ **Data Processing & Analytics**

31. How do you perform data processing in ADLS using Azure Data Factory?
32. What is U-SQL and where is it used?
33. How do you use Apache Spark with ADLS?
34. What is the role of Azure Synapse Analytics with ADLS?
35. How do you optimize performance when querying data in ADLS?

- 36. How does partitioning improve query performance in ADLS?
 - 37. What file formats are best suited for analytics in ADLS?
 - 38. How do you manage metadata for datasets in ADLS?
 - 39. What is Delta Lake and how is it used with ADLS?
 - 40. How would you perform exploratory data analysis on data stored in ADLS?
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♦ **Best Practices & Optimization**

- 41. How do you organize folders and files in ADLS?
 - 42. What are naming conventions best practices for ADLS?
 - 43. How do you ensure high availability and disaster recovery in ADLS?
 - 44. What is the best way to manage large files in ADLS?
 - 45. How do you optimize storage costs in ADLS?
 - 46. How do you manage different environments (dev/test/prod) in ADLS?
 - 47. What are performance tuning tips for working with large datasets in ADLS?
 - 48. How do you enable data versioning in ADLS?
 - 49. How do you implement data governance in ADLS?
 - 50. What is the recommended approach for schema evolution in ADLS?
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♦ **Security & Compliance**

- 51. How do you ensure compliance with GDPR using ADLS?
- 52. How does Azure Purview integrate with ADLS?

- 53. What encryption methods are available in ADLS?
 - 54. Explain customer-managed keys (CMK) vs Microsoft-managed keys (MMK) in ADLS.
 - 55. How do you perform access audits in ADLS?
 - 56. What is data masking and can it be applied in ADLS?
 - 57. How can you anonymize sensitive data in ADLS?
 - 58. How does private endpoint access work with ADLS?
 - 59. How do you configure firewall rules for ADLS?
 - 60. What are secure transfer requirements in ADLS?
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◆ **Monitoring & Troubleshooting**

- 61. How do you monitor performance metrics in ADLS?
 - 62. What are the common issues faced during data ingestion to ADLS?
 - 63. How do you handle schema mismatches in ingestion pipelines?
 - 64. What is Azure Monitor and how does it work with ADLS?
 - 65. How can you enable diagnostic logging in ADLS?
 - 66. What are the troubleshooting steps when files fail to upload?
 - 67. How do you detect and fix performance bottlenecks in ADLS queries?
 - 68. How do you trace permission issues in ADLS?
 - 69. How do you verify if ACLs are correctly set in ADLS?
 - 70. What is the role of Azure Log Analytics with ADLS?
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◆ **Advanced Concepts**

71. How do you implement data lakehouse architecture using ADLS?

72. What is the role of Medallion architecture in data lakes?

73. Explain the concept of time travel in data lakes.

74. What are Change Data Capture (CDC) patterns in ADLS?

75. How do you automate metadata extraction in ADLS?

76. What is hierarchical namespace and why is it important?

77. How do you implement delta processing in ADLS pipelines?

78. How does schema-on-read differ from schema-on-write in ADLS?

79. What is data lake federation?

80. How do you build a data catalog on top of ADLS?

◆ **Scenario-Based Questions**

81. How would you migrate from HDFS to ADLS Gen2?

82. How do you handle real-time log analytics in ADLS?

83. Describe a scenario where you had to enforce security at row level in ADLS.

84. How would you back up and restore data from ADLS?

85. Describe a high-throughput data ingestion pipeline using ADLS.

86. How would you implement a governance strategy for multi-tenant ADLS usage?

87. How do you archive historical data in ADLS?

88. How would you isolate data for different business units in ADLS?

89. How would you implement data validation and error handling in ADLS pipelines?
90. How do you design a resilient pipeline to ADLS using retries and fallbacks?
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◆ **Tools & Ecosystem Integration**

91. How does Power BI connect with ADLS?
92. What is the use of Azure Stream Analytics with ADLS?
93. How do you use Azure Event Grid with ADLS?
94. What are common tools for data wrangling with ADLS?
95. How does Apache NiFi integrate with ADLS?
96. How do you visualize big data from ADLS?
97. How does Microsoft Fabric relate to ADLS?
98. How do you use Azure CLI or PowerShell to manage ADLS?
99. Can you automate ADLS resource provisioning with ARM/Bicep?
100. What is the role of Terraform in managing ADLS infrastructure?